Republic of the Philippines
CEBU TECHNOLOGICAL UNIVERSITY
MAIN CAMPUS
M. J. Cuenco Avenue Cor. R. Palma Street, Cebu City, Philippines
Website: http://www.ctu.edu.ph E-mail: boardsec@ctu.edu.ph, borsec2018@gmail.com
Telephone: 402 4060 loc. 1131

BAGONG PILIPINAS

COLLEGE OF COMPUTER, INFORMATION AND COMMUNICATIONS TECHNOLOGY

CC225 AND CC225L INFORMATION MANAGEMENT

BS INFORMATION TECHNOLOGY

CODSS: COVACHA DIAGNOSTIC SERVICE SYSTEM – INTEGRATED PATIENT REGISTRATION, RECORDS, AND DOCTOR'S DIAGNOSIS

Submitted By:

Bejec, Dan Anton

Limosnero, Jayson Gabriel

Rondina, Roy Adrian

Under the Guidance of:

Gibe S. Tirol, MIT

Faculty, CCICT

1.2 INTRODUCTION

1.2.1 Background of the Study

Traditional paper-based hospital record-keeping systems have long been the standard in healthcare management. However, they present significant challenges that impact efficiency, security, and accessibility.

One of the primary issues with manual record-keeping is limited accessibility and delays in retrieval. Healthcare professionals often struggle to locate and update patient records quickly, which slows down medical processes and treatment decisions. According to Amaechi, Agbasonu, and Nwawudu (2018), paper records are hard to access and take considerable time to update, leading to inefficiencies in hospital workflows.

Another critical concern is data security and loss. Paper records are vulnerable to physical damage, theft, or accidental destruction due to factors such as fire, moisture, or misplacement. Yahaya et al. (2019) note that a major problem with traditional record systems is the risk of records being destroyed, which compromises the availability of essential patient information.

Additionally, difficulty in sharing data between locations hinders coordinated patient care. When a patient transfers from one hospital to another, medical professionals often rely on physical records that need to be manually transported, leading to delays and the risk of missing critical information. Amaechi et al. (2018) emphasize that sharing data between locations is slow, affecting the efficiency of healthcare services.

Traditional systems also contribute to human error and inconsistencies in patient documentation. Handwritten records can be difficult to read and may lead to misinterpretation or incorrect data entry. Yahaya et al. (2019) highlight how automation reduces paperwork and standardizes data input, eliminating errors associated with manual entry.

Furthermore, long-term storage challenges make maintaining accurate medical histories difficult. Paper-based files require large physical storage spaces, and as hospital archives expand, it becomes increasingly difficult to organize and retrieve records. Amaechi et al. (2018) warn that keeping records for extended periods increases the risk of damage or loss, making medical history retrieval unreliable.

The implementation of electronic healthcare record management systems addresses these concerns by enhancing accessibility, security, and data integration. Automated systems streamline patient record retrieval, ensure secure data encryption, and allow seamless sharing between healthcare facilities, ultimately improving patient care efficiency.

1.2.2 Statement of the Problem

The Covacha Diagnostic Services Clinic currently relies on a manual, paper-based record-keeping system that creates significant inefficiencies in patient data management. The lack of a structured digital system leads to service delays, an increased risk of errors, and difficulty retrieving critical patient information, ultimately negatively impacting both medical staff and patients. Given these operational challenges, the clinic requires a digital solution that will improve record management through a secure, centralized database while automating tasks to enhance efficiency and data security.

The proposed system aims to:

1. Develop a system that the admin can:
    - 1.1. View a dashboard overview of clinic operations.
    - 1.2. Manage doctor, staff, and patient records.
    - 1.3. Add/modify doctor and staff details.
    - 1.4. Configure laboratory tests and payment values.
    - 1.5. Oversee patient and laboratory test management.
2. Develop a system that a doctor can:
    - 2.1. Access a dashboard for quick patient insights.

2.2. View and update patient medical records.

2.3. Add diagnoses and laboratory test requests.

2.4. Oversee patient medical history.

3. Develop a system that the staff can:

3.1. Monitor daily operations via an interactive dashboard.

3.2. Register and update patient check-up details.

3.3. Process laboratory tests and transactions efficiently.

By achieving these objectives, CODSS will eliminate manual inefficiencies, reduce human errors, and ensure seamless coordination between doctors, staff, and patients - ultimately delivering faster, more reliable healthcare services.

1.2.3 Objectives

The primary objective of CODSS (Covacha Diagnostic Service System) is to develop an advanced record management system that provides clinic staff and administrators with a secure, centralized digital platform. The system aims to automate critical workflows to improve operational efficiency while implementing robust security measures to protect sensitive patient data.

Specifically, it aims to:

- Develop a clinic management platform utilizing Python, PyQt, and PostgreSQL to optimize patient onboarding, electronic health records, and clinical assessment workflows
- Design an optimized database to centralize patient, doctor, and staff records while minimizing data redundancy and ensuring integrity.
- Enhance operational efficiency by automating manual processes to reduce service delays and human errors.
- Ensure data security through encrypted storage and secure access protocols for sensitive medical information.
- Improve patient care delivery by enabling real-time access to complete medical records for accurate diagnosis and treatment.

1.2.4 Significance of the Study

This system will be beneficial as it will serve as an efficient tool for managing patient records, improving data accuracy, and enhancing the overall workflow of the clinic. By automating key processes, the system will reduce administrative workload and minimize the risk of misplaced or lost records.

Moreover, this system can be beneficial to the following:

- **Patients** - The system will enable faster registration and medical record retrieval, resulting in quicker diagnoses and more accurate treatments.
- **Doctors** - It will provide immediate access to complete patient histories and diagnostic reports, supporting better clinical decision-making and more efficient care delivery.
- **Clinic Staff** - The system will streamline administrative tasks, reduce workload, and facilitate more organized management of patient information.
- **Clinic Owner** - It will improve overall operations by enhancing record management, minimizing errors, and ensuring robust data security.
- **Future Developers** - They will benefit from the system's modular architecture, comprehensive documentation, and flexible design that enables seamless upgrades and integration with future healthcare technologies.

Ultimately, this digital transformation will modernize clinic operations, elevate service quality, and establish a foundation for sustainable growth in an increasingly digital healthcare landscape.

1.3 SCOPE AND LIMITATION

1.3.1 Scope

The proposed Clinic Diagnostic System is a specialized digital platform designed exclusively for private healthcare clinics to revolutionize how patient records and critical medical documents are managed. This comprehensive system will be utilized by all key clinic personnel, including physicians for patient care, administrative staff for daily operations, and clinic owners for oversight and decision-making. Its core functionality will focus on:

- **Patient Record Management** - The system will maintain complete patient profiles, including medical histories and visit records, in a secure digital database for organized storage and quick access.
- **Diagnosis and Document Printing** - It will generate and print essential medical documents like e-prescriptions and lab reports with standardized formats for accuracy and efficiency.
- **Data Retrieval** - The system enables instant patient record access through search filters (ID, name, date) to support consultations and follow-ups.

This system aims to lessen administrative tasks, reduce paperwork, and improve the overall workflow of the clinic.

1.3.2 Limitations

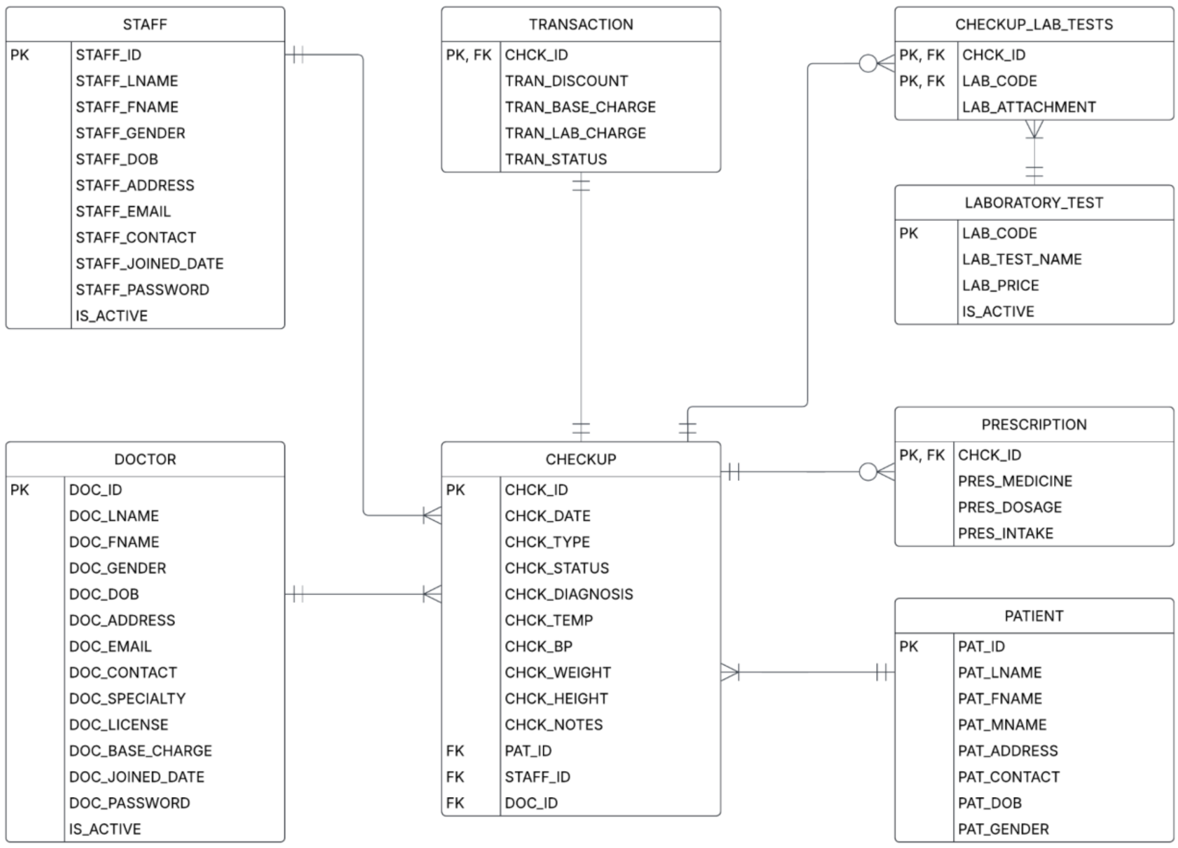
While the system significantly enhances record-keeping and document management capabilities, it operates within specific constraints. The solution is tailored to the clinic's current operational needs and technological infrastructure. These intentional design boundaries result in the following limitations:

- **Limited to a Single Private Clinic** - The system is exclusively designed for the client's standalone clinic operation. It does not support multi-branch implementations or integration with other healthcare facilities.
- **No Remote Access** - All system functions are restricted to in-clinic usage through local workstations. Remote access capabilities or cloud-based functionality are not included in this implementation.
- **No Online Appointment Scheduling** - Appointment management remains a staff-mediated process. Patients cannot self-schedule or modify appointments through the digital platform.
- **No Billing or Payment Processing** - The system deliberately omits payment processing and billing functions. Insurance claims management and financial transactions fall outside the project scope.

1.4 METHODOLOGY

1.4.1 Database Design

1.4.1.1 Entity-Relationship Diagram (ERD) Conceptual Model



1.4.1.2 Data Dictionary

Table Name	Attribute Name	Contents	Type	Format	Range	Required	PK or FK	FK Referenced Table
CHECKUP	CHCK_ID	Unique Identifier for each checkup	VARCHAR(30)	Xxxxxx-xxx	N/A	Y	PK	
	CHCK_DATE	Date of the Checkup	DATE	YYYY-MM-DD	N/A	Y		
	CHCK_TYPE	Type of Checkup	VARCHAR(50)	Xxxxxxxxx	New Checkup, Follow-Up Checkup			
	CHCK_STATUS	Status of the checkup	VARCHAR(20)	Xxxxxxxxx	pending, ongoing, complete,			
	CHCK_BP	Blood pressure reading	VARCHAR(20)	Xxxxxxxxx	N/A	Y		
	CHCK_HEIGHT	Height measurement	VARCHAR(20)	Xxxxxxxxx	N/A	Y		
	CHCK_WEIGHT	Weight measurement	VARCHAR(20)	Xxxxxxxxx	N/A	Y		

	CHCK_TEMP	Body temperature	VARCHAR(50)	Xxxxxxxxx	N/A	Y		
	CHCK_DIAGNOSIS	Diagnoses or conditions noted during the checkup	VARCHAR(250)	Xxxxxxxxx	N/A			
	CHCK_NOTES	Additional notes or observations	VARCHAR(100)	Xxxxxxx	N/A			
	PAT_ID	Identifier for the patient	INT	#########	100000000 - 99999999	Y	FK	PATIENT
	STAFF_ID	Identifier for the staff member	INT	######	100000 - 999999	Y	FK	STAFF
	DOC_ID	Identifier for the doctor	INT	#####	10000-99999		FK	DOCTOR
DOCTOR	DOC_ID	Doctor Unique identifier for each doctor	INT	#####	10000-99999	Y	PK	
	DOC_LNAME	Doctor Last Name	VARCHAR(50)	Xxxxxxxxxx	N/A	Y		
	DOC_FNAME	Doctor First Name	VARCHAR(50)	Xxxxxxxxx	N/A	Y		

	DOC_MNAME	Doctor Middle Name	VARCHAR(50)	Xxxxxxxxx	N/A			
	DOC_GENDER	Gender of the doctor	VARCHAR(20)	Xxxxxxxxxxx	Male, Female	Y		
	DOC_DOB	Doctor's date of birth	DATE	YYYY-MM-DD	N/A	Y		
	DOC_ADDRESS	Doctor Address	VARCHAR(50)		N/A	Y		
	DOC_CONTACT	Doctor Contact Number	VARCHAR(50)	Xxxxxxxxxxx	N/A	Y		
	DOC_EMAIL	Doctor Email	VARCHAR(50)	Xxxxxxxxxxx	N/A	Y		
	DOC_SPECIALTY	Area of specialization	VARCHAR(50)	Xxxxxxxxxxx	N/A	Y		
	DOC_LICENSE	Medical license number	INT			Y		
	DOC_RATE	Doctor Rate	INT					
	DOC_PASSWORD	Encrypted password	VARCHAR(200)	Xxxxxxxxxxx	N/A	Y		
	DOC_JOINED_DATE	Date when the doctor joined the	DATE	YYYY-MM-DD	N/A	Y		

		clinic						
	IS_ACTIVE	Indicates if the test is currently available	BOOLEAN	True/False	true/false (deafult : true)			
LABORATORY_TEST	LAB_CODE	Laboratory Code	VARCHAR(20)	Xxxxxxxxxxx	N/A	Y	PK	
	LAB_TEST_NAME	Name of Laboratory Test	VARCHAR(20)	Xxxxxxxxxxx	N/A	Y		
	LAB_PRICE	Laboratory Price	INT	####	0-9999	Y		
	IS_ACTIVE	Indicates if the test is currently available	BOOLEAN	True/False	true/false (default: true)			
PATIENT	PAT_ID	Patient ID	INT			Y	PK	
	PAT_LNAME	Patient Last Name	VARCHAR(50)	Xxxxxxxxxx	N/A	Y		

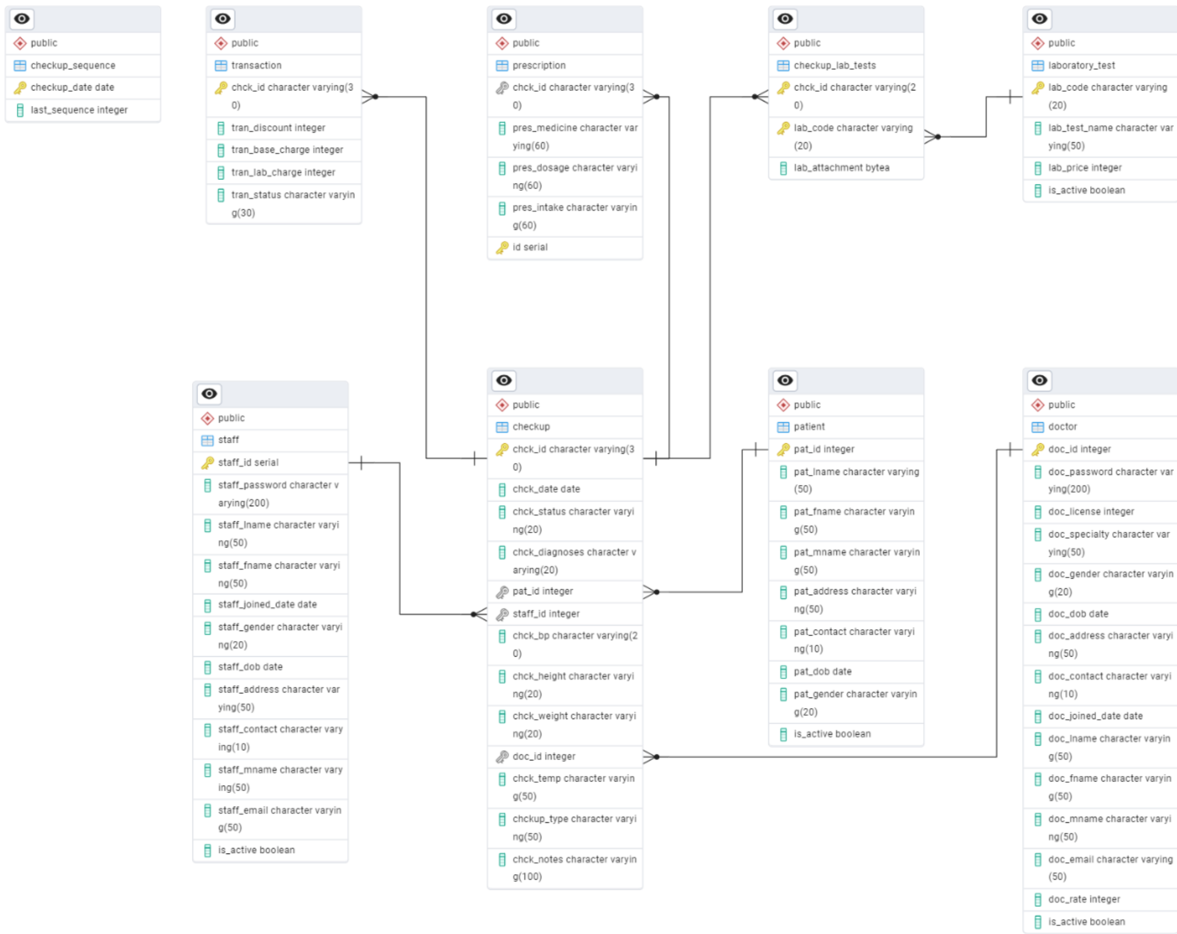
	PAT_FNAME	Patient First Name	VARCHAR(50)	Xxxxxxxxxx	N/A	Y		
	PAT_MNAME	Patient Middle Name	VARCHAR(50)	Xxxxxxxxxx	N/A			
	PAT_ADDRESS	Patient Address	VARCHAR(50)	Xxxxxxxxxx	N/A	Y		
	PAT_CONTACT	Patient Contact Number	VARCHAR(10)	Xxxxxxxxxx		Y		
	PAT_DOB	Patient Date of Birth	DATE	YYYY-MM-DD	N/A	Y		
	PAT_GENDER	Patient Gender	VARCHAR(20)	Xxxxxx	Male/ Female	Y		
	IS_ACTIVE	Indicates if the patient is currently available	BOOLEAN	True/False	true/false (default: true)			
PRESCRIPTION	ID	Unique identifier for each prescription record	INTEGER	###xxxxxxxxx	0-99	Y	PK	
	PRES_MEDICINE	Prescription Medicine	VARCHAR(60)	Xxxxxxxxxx	N/A	Y		

	PRES_DOSAGE	Prescription Dosage	VARCHAR(60)	Xxxxxxxxxx	N/A	Y		
	PRES_INTAKE	Prescription Intake	VARCHAR(60)	Xxxxxxxxxx	N/A	Y		
	CHCK_ID	Checkup ID	VARCHAR(30)	XXXXXX-XXX	N/A	Y	FK	CHECKUP
STAFF	STAFF_ID	Identifier for the staff member	INT	######	100000 - 999999	Y	PK	
	STAFF_PASSWORD	Staff Password	VARCHAR(200)	XxxxxxxxXxxxx xxxxx	N/A	Y		
	STAFF_LNAME	Staff Last Name	VARCHAR(50)	XxxxxxxxXxxxx xxxxx	N/A	Y		
	STAFF_FNAME	Staff First Name	VARCHAR(50)	XxxxxxxxXxxxx xxxxx	N/A	Y		
	STAFF_JOINED_DAT E	Date Staff Joined	DATE	YYYY-MM-DD	N/A	Y		
	STAFF_GENDER	Staff Gender	VARCHAR(20)	XxxxxxXxxxxxx xxx	Male/ Female	Y		
	STAFF_DOB	Staff Date of Birth	DATE	YYYY-MM-DD	N/A	Y		
	STAFF_ADDRESS	Staff Address	VARCHAR(50)	XxxxxxxXxxxxx	N/A	Y		

				xxxx				
	STAFF_CONTACT	Staff Contact Number	VARCHAR(10)	XXXXXXXXXxx xxxxxxx	N/A	Y		
	STAFF_MNAME	Staff Middle Name	VARCHAR(50)	XxxxxxxXxxxxx xxxx	N/A	Y		
	STAFF_EMAIL	Staff Email	VARCHAR(50)	XxxxxxxxXxxxx xxxxx	N/A	Y		
	IS_ACTIVE	Indicates if the staff is currently available	BOOLEAN	True/False	true/false (default:true)	Y		
TRANSACTION	CHCK_ID	Checkup ID	VARCHAR(30)	Xxxxxxxxxx	N/A	Y	PK, FK	CHECKUP
	TRAN_DISCOUNT	Transaction Discount	INT					
	TRAN_BASE_CHARGE	Transaction Base Charge	INT					
	TRAN_LAB_CHARGE	Transaction Laboratory Charge	INT					

	TRAN_STATUS	Transaction Status	VARCHAR(30)	Xxxxxxxxxx	N/A			
CHECKUP_LAB_TESTS	CHCK_ID	Checkup ID	VARCHAR(20)	XXXXXX-XXX	N/A	Y	PK, FK	CHECKUP
	LAB_CODE	Laboratory Code	VARCHAR(20)	XXX-XXX	N/A	Y	PK, FK	LABORATOR Y
	LAB_ATTACHMENT	Laboratory Attachment	BYTEA		N/A	Y		
CHECKUP_SEQUENCE	CHECKUP_DATE	Date of Checkup	DATE	YYYY-MM-DD	N/A	Y	PK	
	LAST_SEQUENCE	Last Checkup	INT	###	0-99	Y		

1.4.1.3 Entity-Relationship Diagram (ERD) Physical Model



1.5 DEVELOPMENT TOOLS AND TECHNOLOGIES

1.5.1 RDMBS: PostgreSQL

For the relational database management system (RDBMS), **PostgreSQL** was selected as the relational database management system (RDBMS) because of its advanced capabilities, dependability, and strong compliance with SQL standards. As a robust open-source object-relational database, it ensures ACID compliance, maintains high data integrity, and supports complex queries and diverse data types. Its extensible architecture allows the addition of various features through extensions, making it suitable for a wide range of data storage requirements. Additionally, its well-established community and thorough documentation offer valuable resources for development and issue resolution.

1.5.2 Programming Language

1.5.2.1 Python

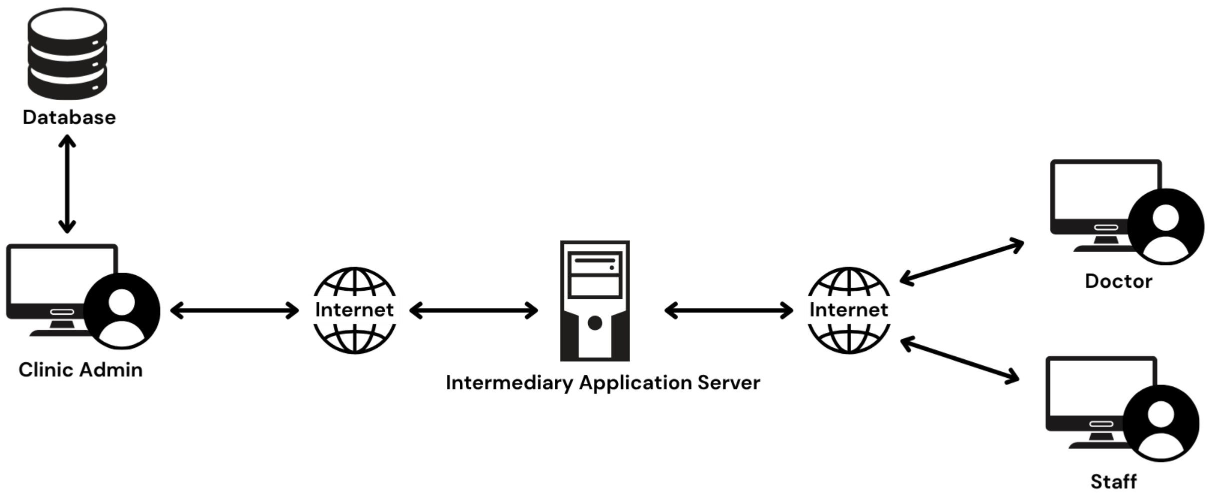
The system utilizes a combination of backend and frontend technologies to create a desktop application that provides a responsive and dynamic user experience.

The system employs **Python** as the primary programming language for desktop application development. Its support for object-oriented programming (OOP) enables modular, maintainable, and scalable code. Python's readability and extensive libraries make it ideal for handling backend logic and data processing efficiently.

1.5.2.2 GUI Framework: Qt Designer

The graphical user interface (GUI) of the application is designed using **Qt Designer**, a drag-and-drop interface builder for the Qt framework. Qt Designer, integrated with Python via the PyQt or PySide libraries, allows for the creation of intuitive and visually appealing user interfaces. This combination ensures a responsive and user-friendly experience across different desktop environments.

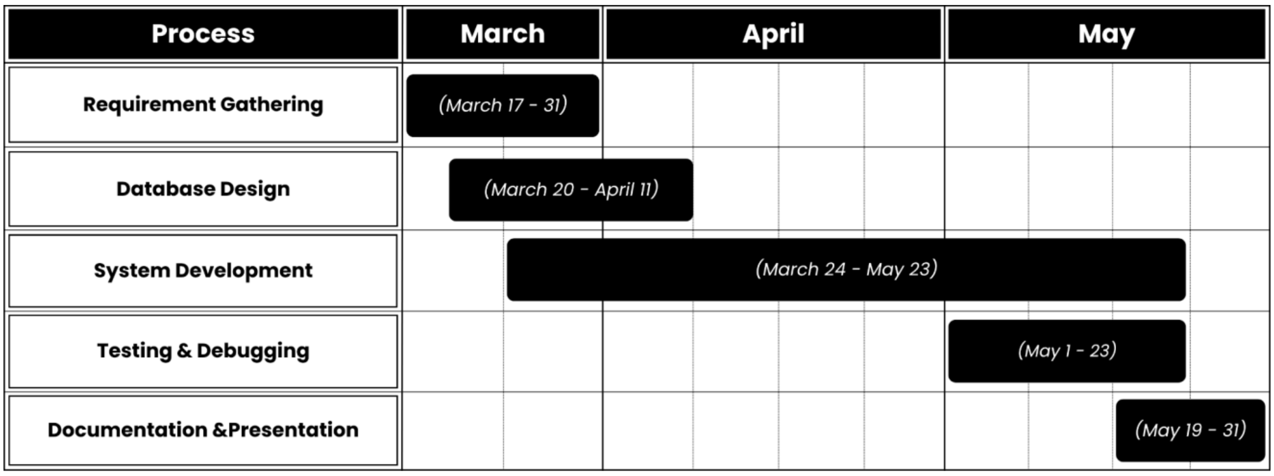
1.5.2.3 System Architecture



The system architecture follows a client-server model with a centralized database that serves multiple user roles, including clinic administrators, doctors, and staff. The intermediary application server acts as the middleware between end-users and the database, facilitating secure data exchange over internet connections. This architecture enables remote access for authorized personnel while maintaining data integrity and security across the clinic management system.

1.6 PROJECT TIMELINE

1.6.1 Gantt Chart



The project schedule outlines the major phases and their timelines from March through May, encompassing approximately eleven weeks in total. The first stage, "Requirement Gathering," runs from March 17 to 31, focusing on detailed planning and client discussions to establish the core needs of CoDSS (Covacha Diagnostic Service System). Simultaneously, the "Database Design" phase begins on March 20 and wraps up by April 11, involving the development of the Entity-Relationship Diagram (ERD) and ensuring a solid data foundation for clinic operations.

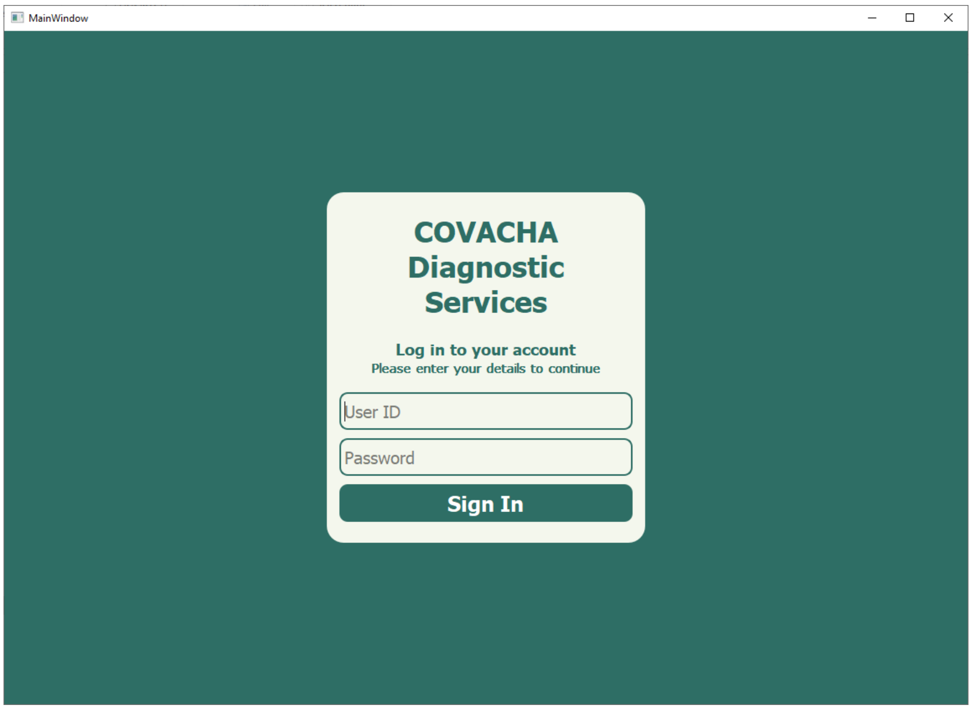
The longest and most intensive phase, "System Development," starts on March 24 and continues until May 23, covering the coding and integration of key features such as patient records, diagnostic processes, and financial transactions. Running alongside the later stages of development, "Testing & Debugging" takes place from May 1 to 23, ensuring system stability and accuracy, especially crucial for healthcare functions like prescription tracking and billing. Finally, the project concludes with "Documentation & Presentation" from May 19 to 31, producing user guides and final presentations to facilitate smooth system adoption.

This structured approach prioritizes early database setup, continuous testing, and thorough documentation, essential for a reliable and user-friendly clinic management system.

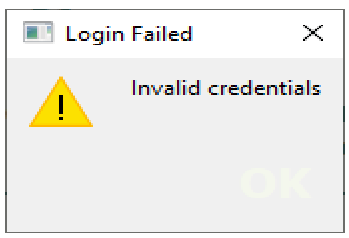
1.7 EXPECTED OUTPUTS

1.7.1 Screenshots of the Program

1.7.1.1 User Login

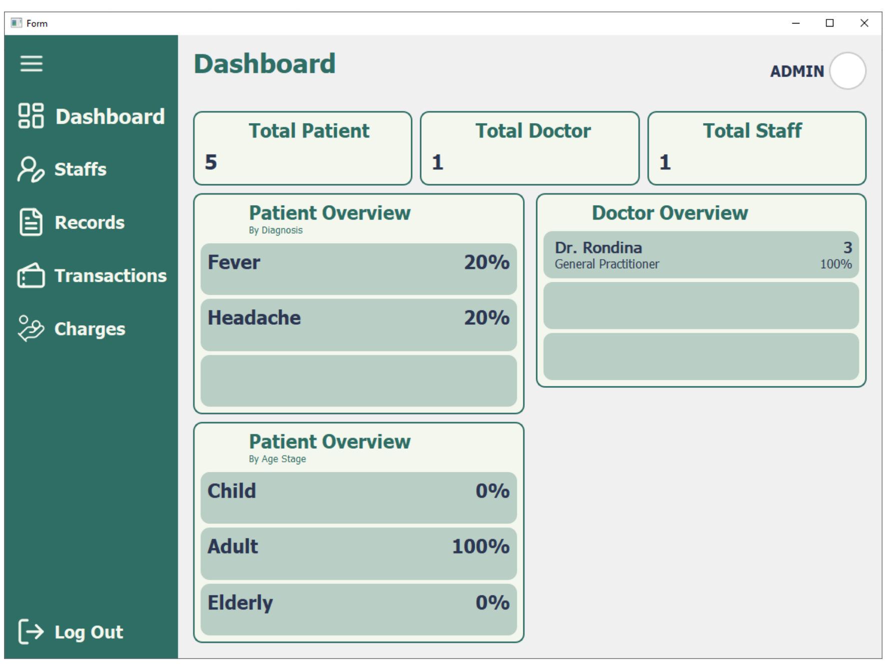


Validation error

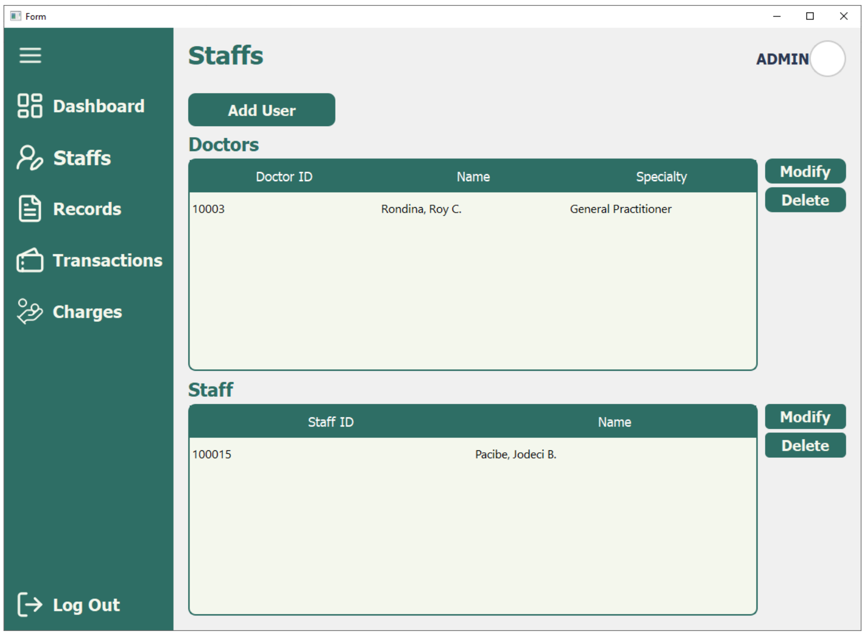


1.7.1.2 Administrator Interface

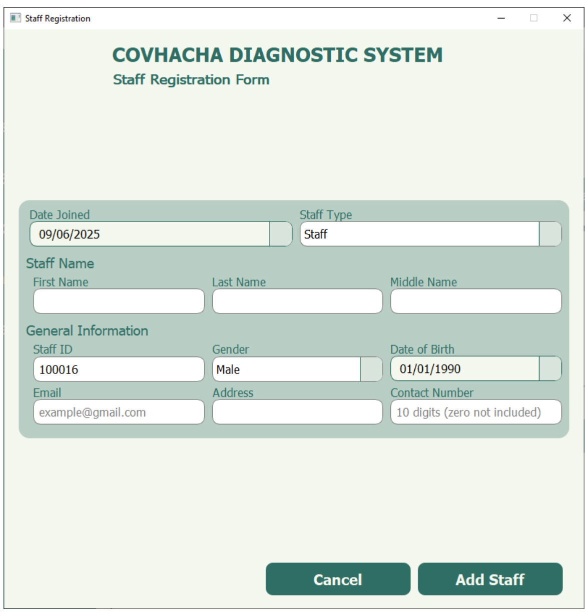
1.7.1.2.1 Dashboard



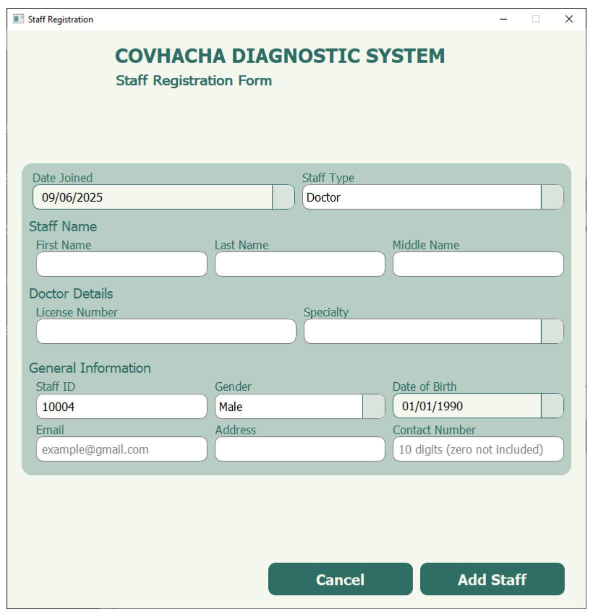
1.7.1.2.2 Staff Management



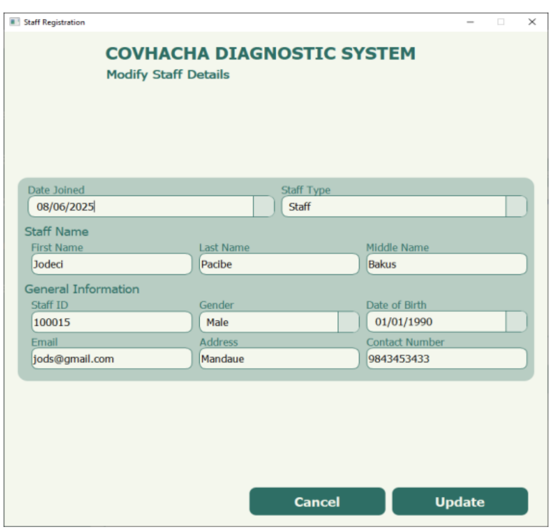
Add Clinic Staff Form



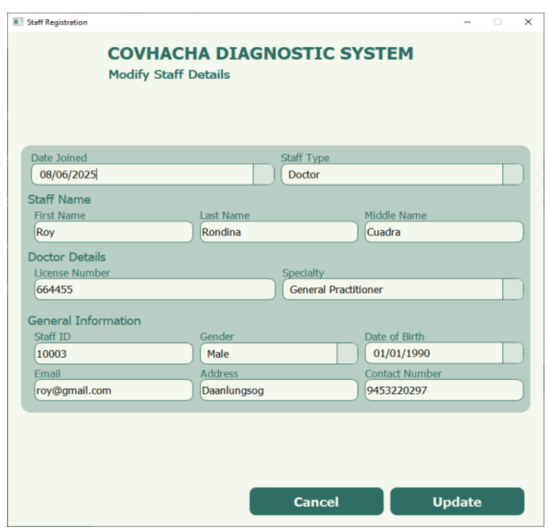
Add Doctor Form



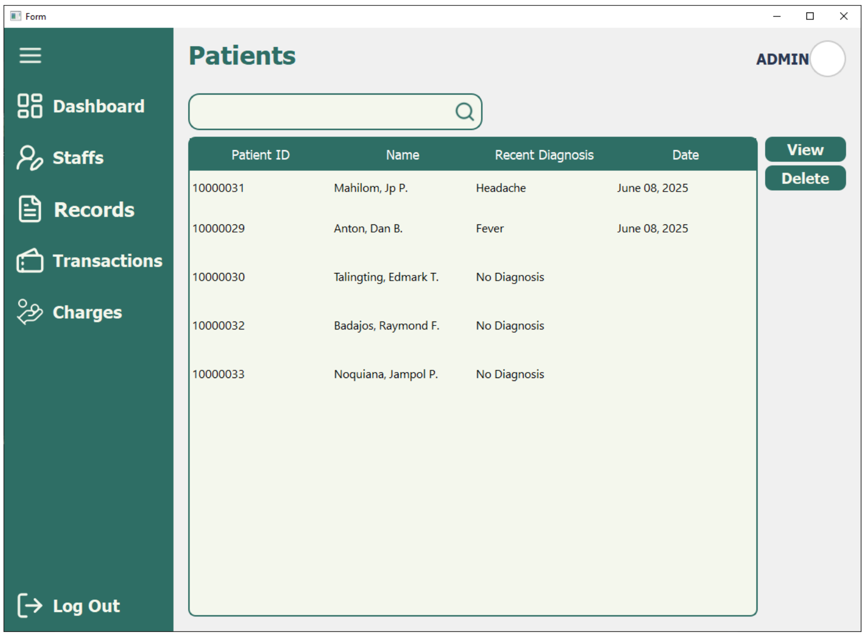
Modify Clinic Staff Form



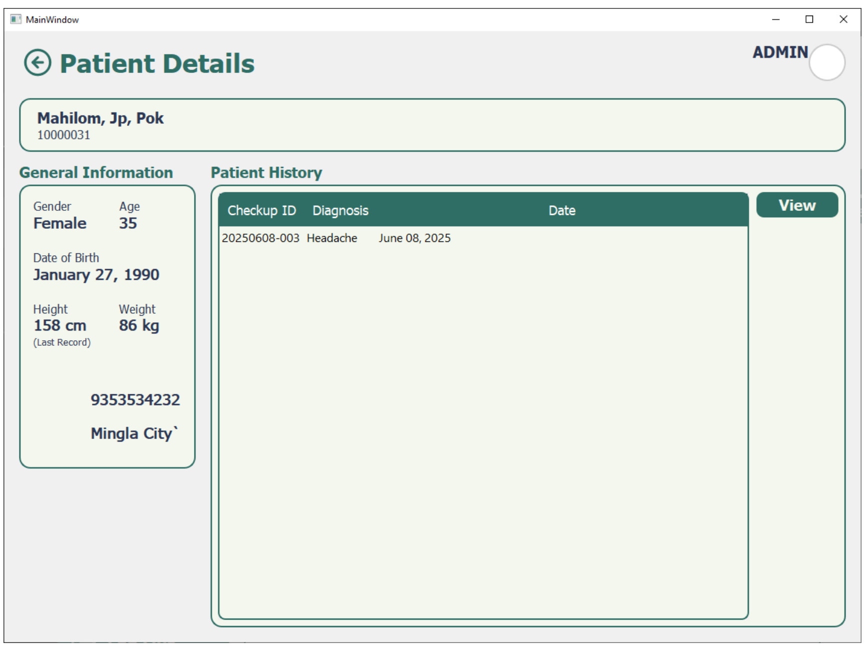
Modify Doctor Form



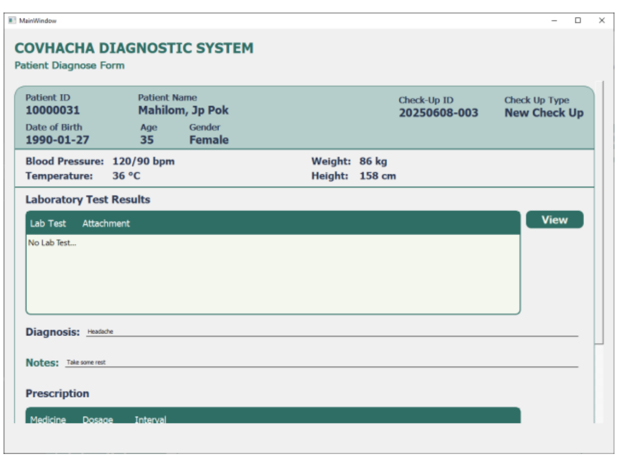
1.7.1.2.3 Patient Overview

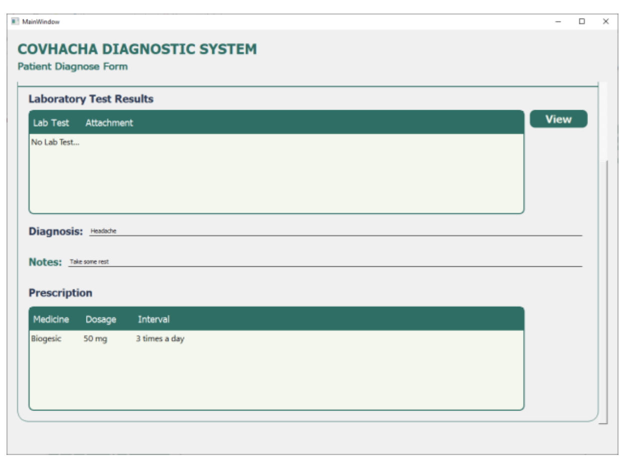


View Patient Details

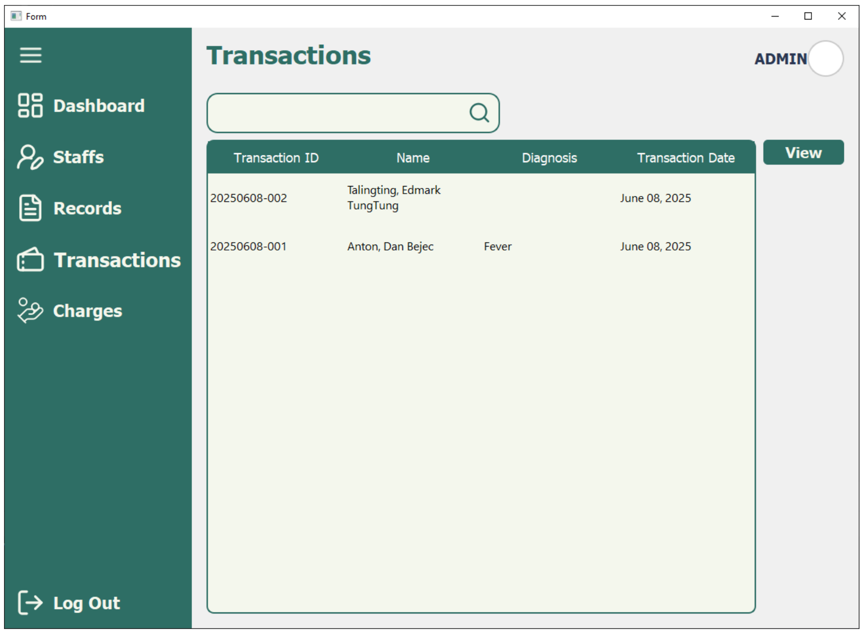


View Patient Diagnosis

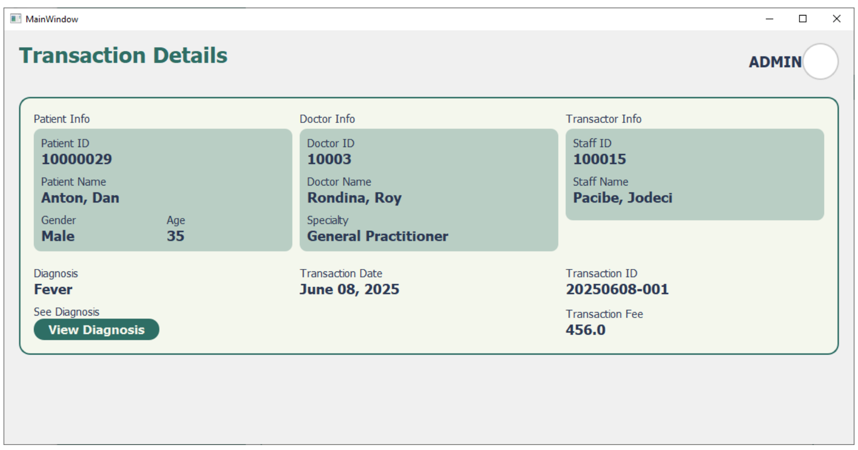




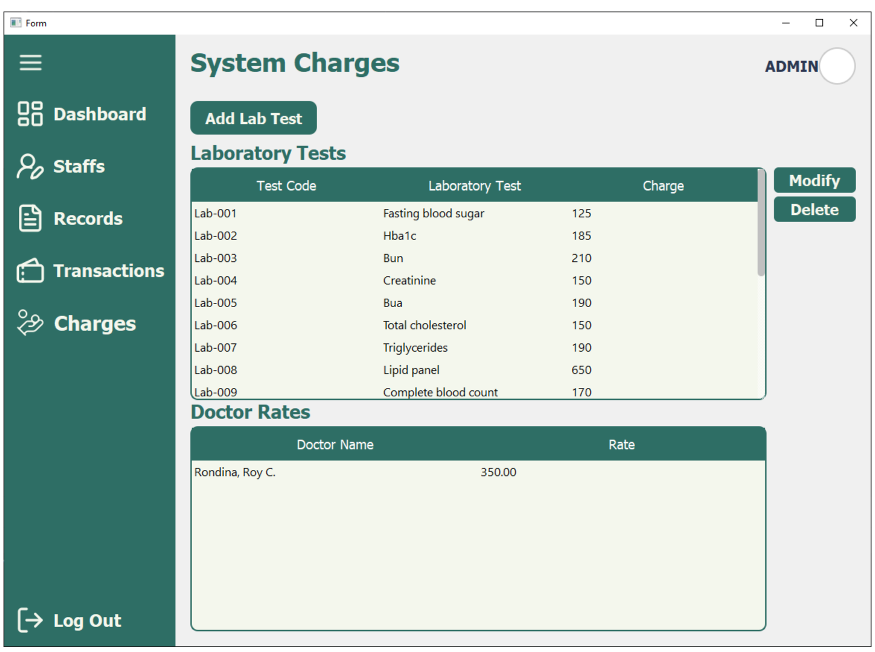
1.7.1.2.3 Transaction Overview



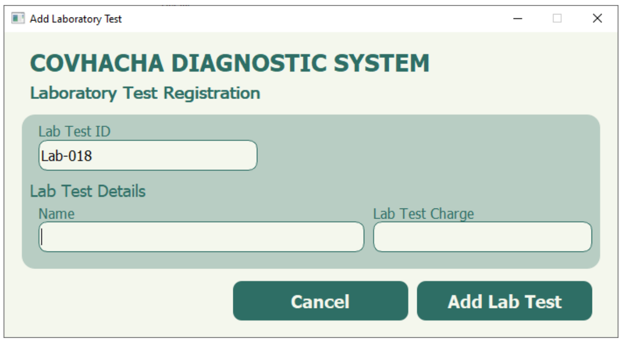
Transaction Details



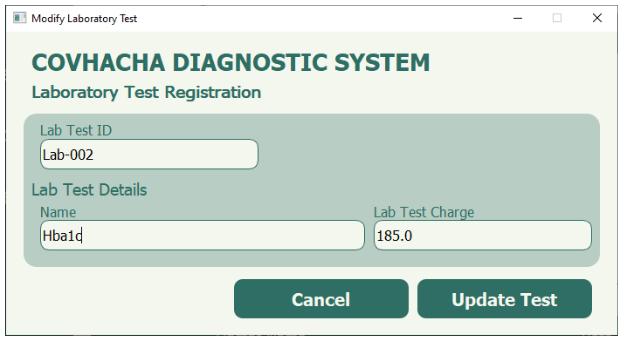
1.7.1.2.3 System Charges Management



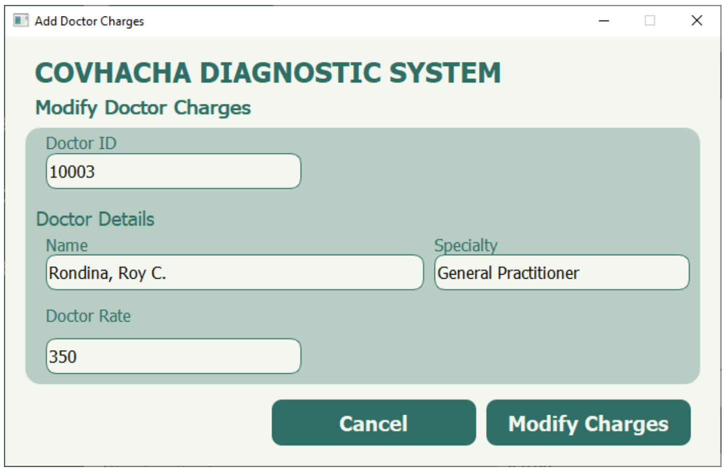
Add Laboratory Test Form



Modify Laboratory Test Form

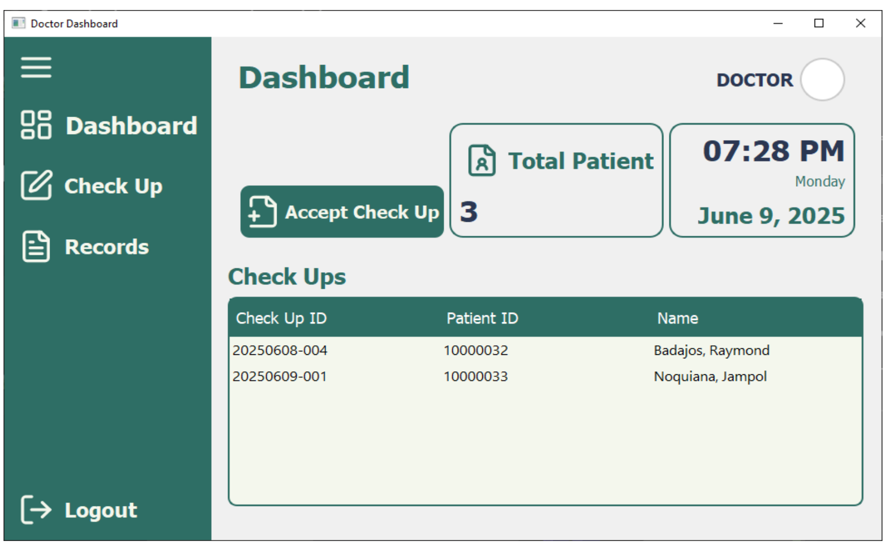


Modify Doctor Charge Form

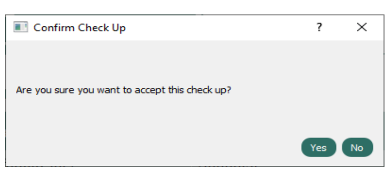


1.7.1.3 Doctor Interface

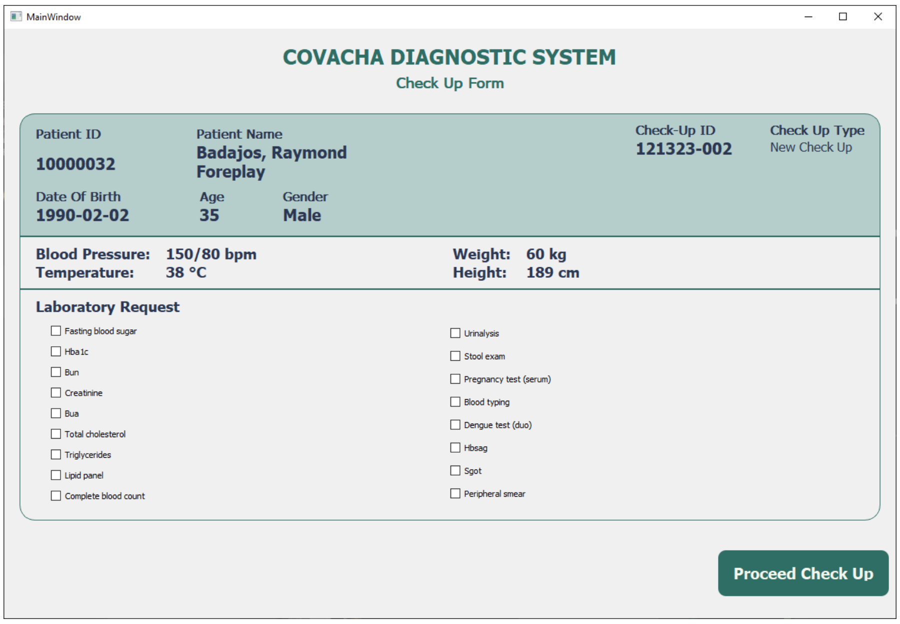
1.7.1.3.1 Dashboard



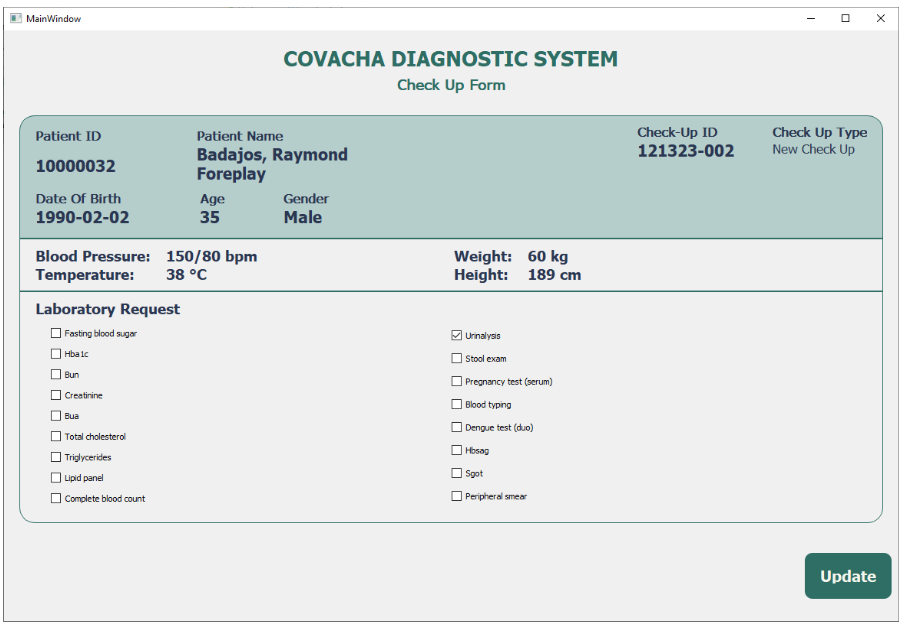
Accept checkup validation



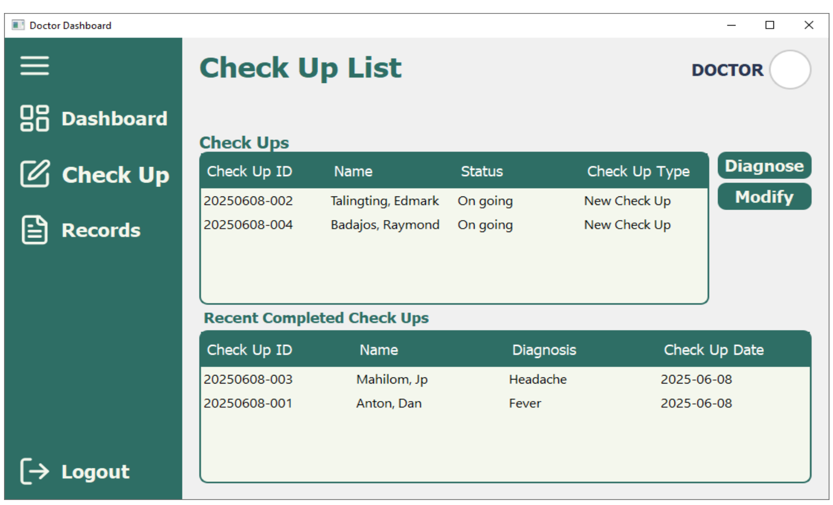
Add Laboratory Request



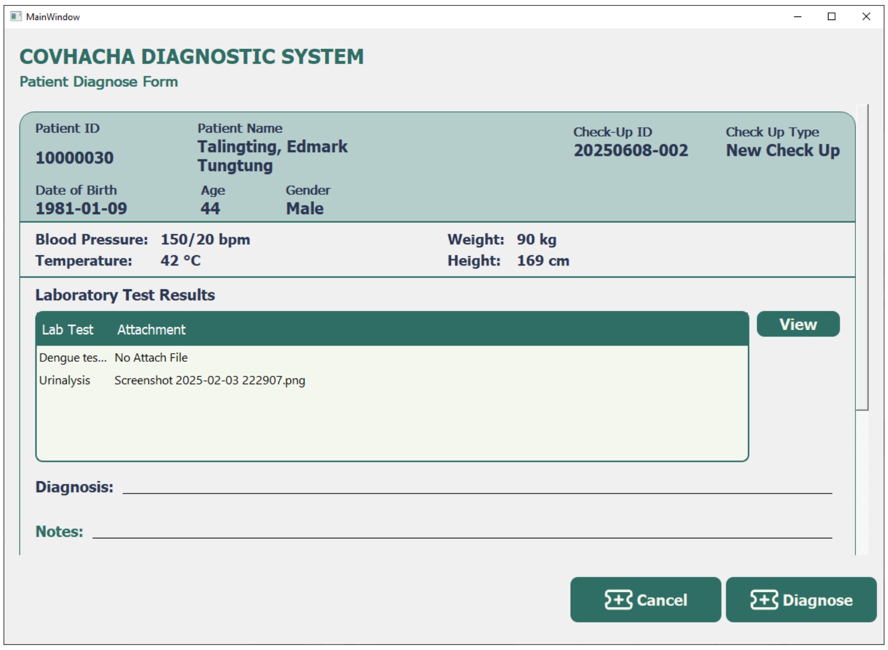
Modify Laboratory Request



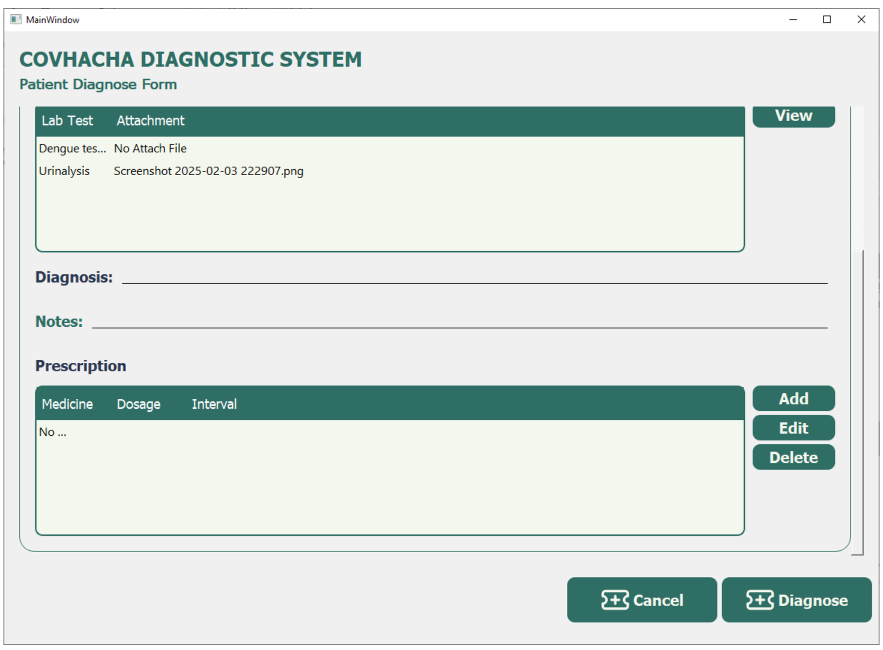
1.7.1.3.1 Checkup List



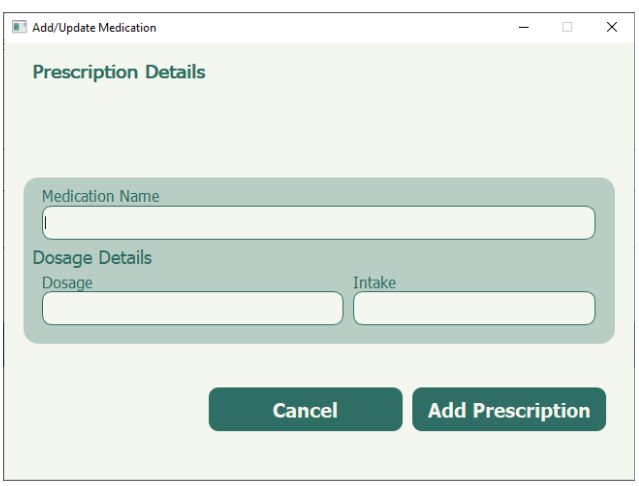
Diagnose Patient



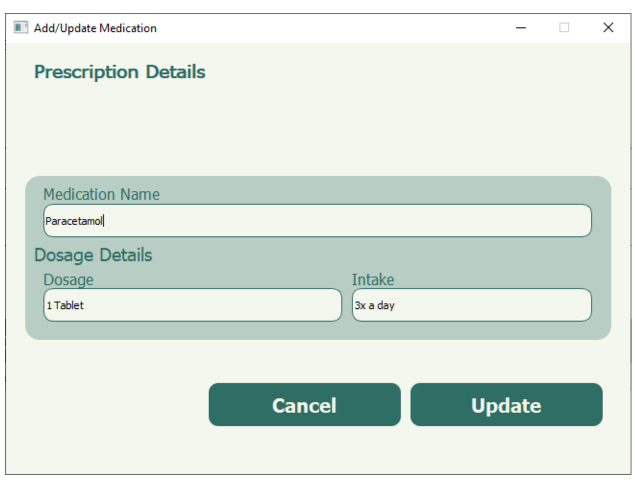
Add Diagnosis and Prescription



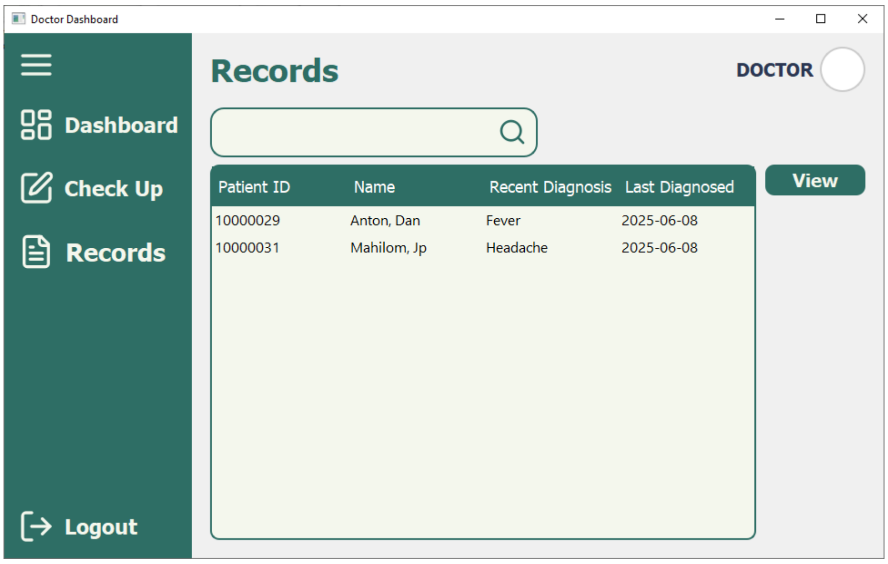
Add Prescription Form



Edit Prescription Form

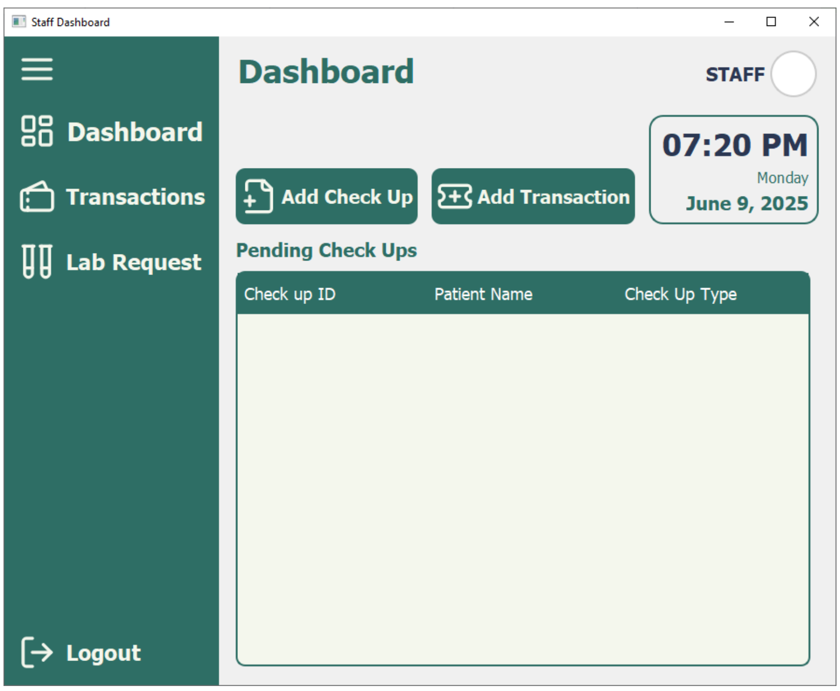


1.7.1.3.1 Doctor's Patient Records

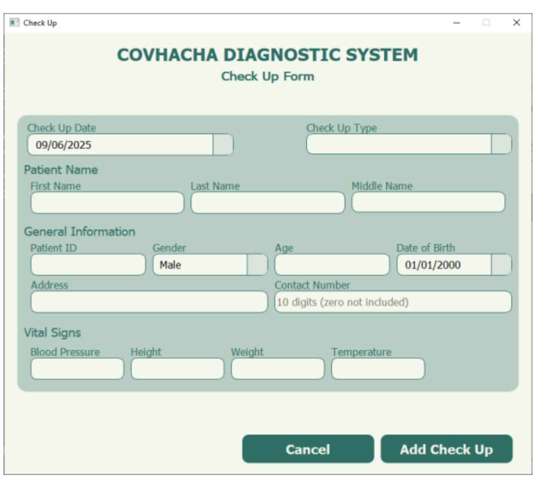


1.7.1.4 Staff Interface

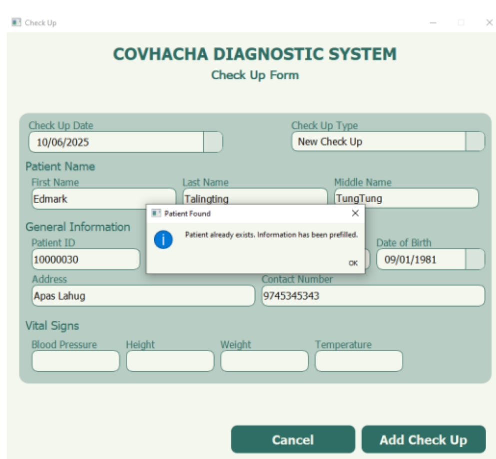
1.7.1.4.1 Dashboard



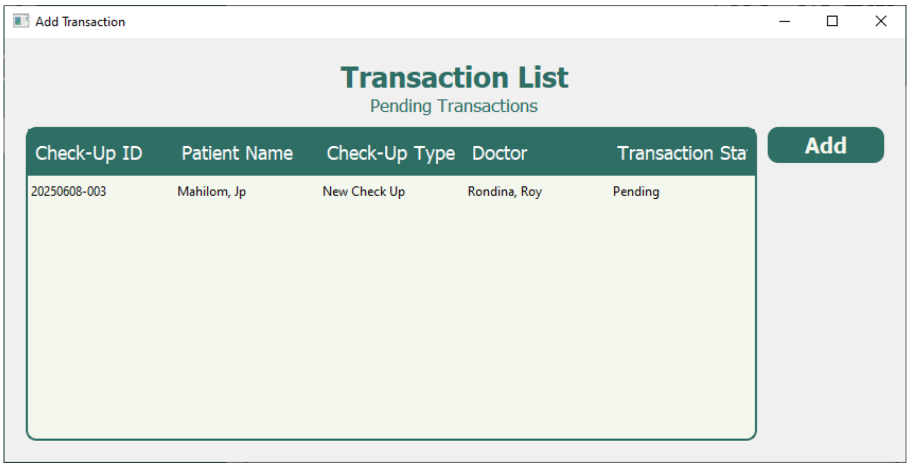
Add Checkup Form



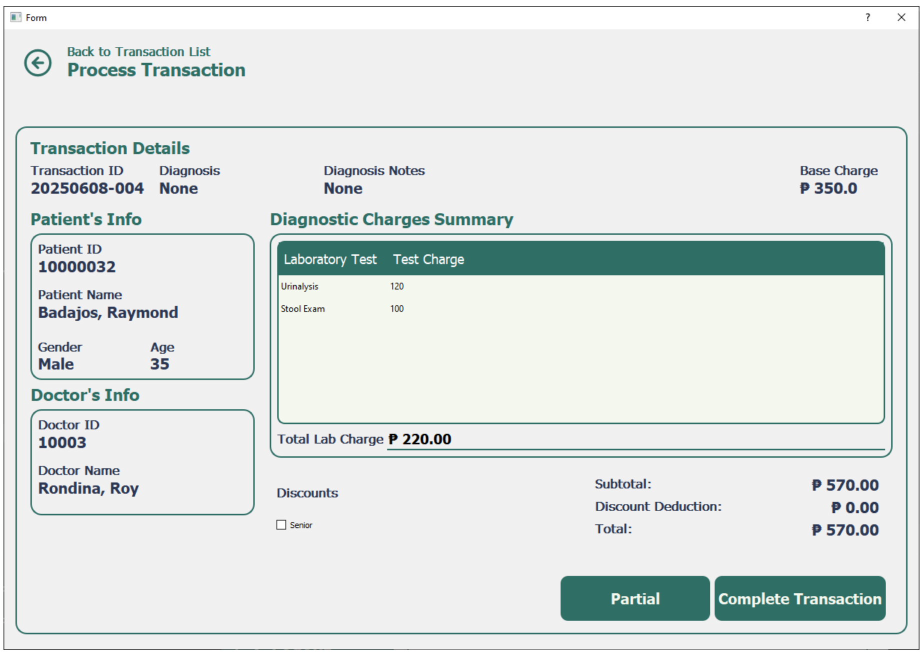
Checkup Form (with existing record)



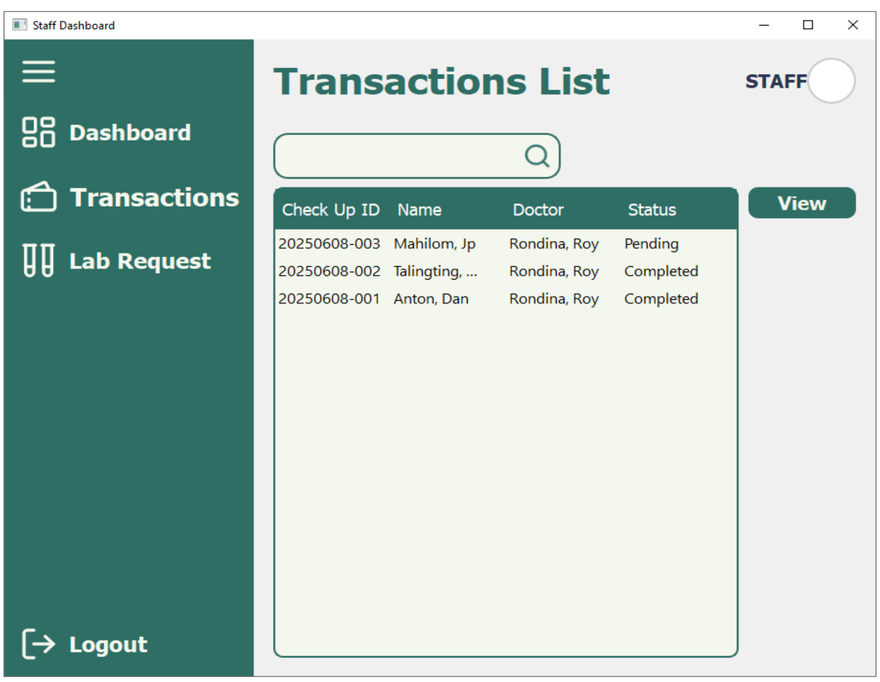
Pending Transaction List



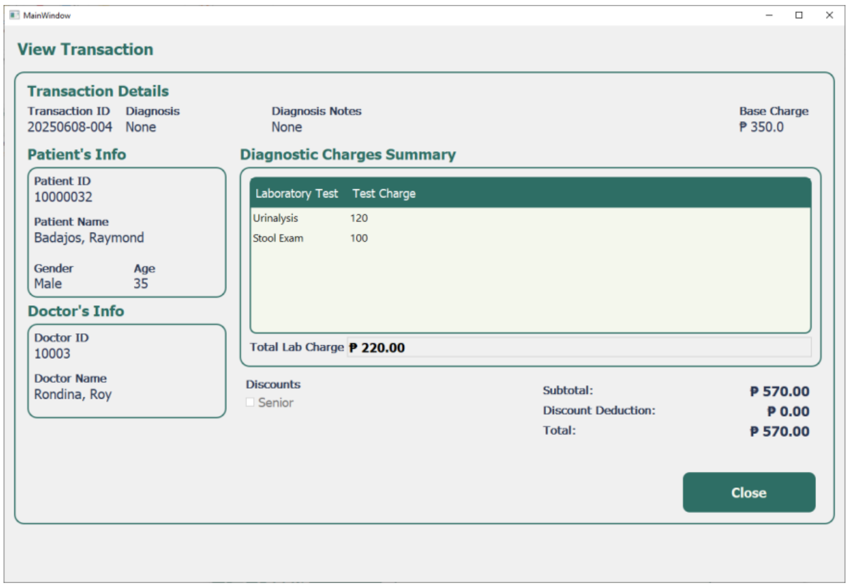
Process Transaction



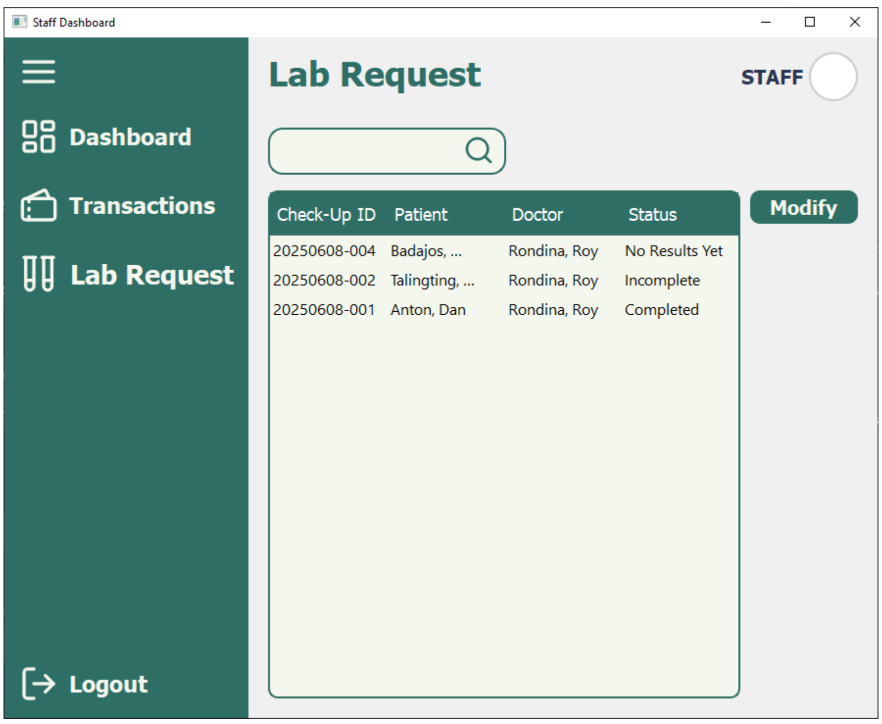
1.7.1.4.2 Transaction List



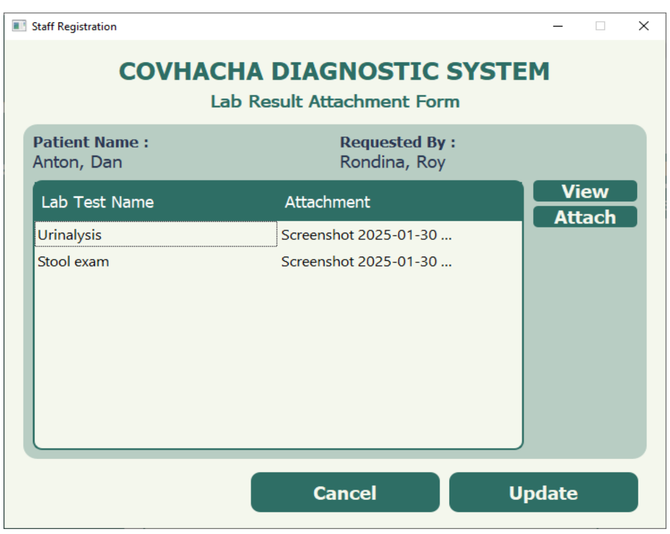
Staff View Transaction Details



1.7.1.4.3 Lab Request List



Add Lab Attachment Form



1.8 CONCLUSION

1.8.1 Summary

The Covacha Diagnostic Services Clinic currently uses a manual, paper-based system for patient records, causing inefficiencies, delays, and errors that affect both staff and patients. The proposed Covacha Diagnostic Service System (CODSS) aims to replace this with a secure, centralized digital platform to automate workflows, improve data accuracy, and enhance operational efficiency. The system will allow admins to manage clinic operations and staff records, doctors to access and update patient medical histories, and staff to efficiently handle patient check-ups and lab tests. CODSS will be developed using Python, PyQt, and PostgreSQL. Focusing on secure data storage, reduced redundancy, and real-time access to patient information. The system will benefit patients through faster service, doctors through better clinical decision-making, staff by reducing administrative workload, and clinic owners by improving overall management.

Implementing CODSS will modernize the Covacha Diagnostic Service Clinic by eliminating manual record-keeping inefficiencies and minimizing human errors. This digital transformation will streamline clinic workflows, enhance data security, and improve patient care quality by providing timely and accurate medical information. While limited in scope to a single clinic and lacking some advanced features like remote access, CODSS lays a strong foundation for future upgrades and integration with broader healthcare technologies, ultimately supporting sustainable growth and improved healthcare delivery.

1.9 REFERENCES

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